HONOURS PROJECT · ARCHITECTURE

Does Skill-Level Inference Improve Resume-to-Job Matching?

An Evidence-Grounded Evaluation of Large Language Models

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THE PROBLEM

Resumes rarely list every skill

Keyword search misses people

A resume lists Python, scikit-learn and predictive modelling, yet never writes “machine learning”. Keyword matching ranks this candidate too low.

LLMs add new risks

LLMs infer unstated skills, but they also invent them. GPT-4 ratings of 736 resumes showed only a minor correlation with human ratings (Vaishampayan et al., 2025).

THE MAIN QUESTION

Does evidence-supported LLM skill inference improve resume-to-job matching beyond semantic matching and explicit skill matching?

LLM inference is treated as an intervention in ranking, tested under control, with every inferred skill checked against the resume.

RESEARCH QUESTIONS

Four questions guide the work

RQ1

How accurately does an LLM infer skills a resume supports but does not state?

RQ2

Does adding inferred skills improve candidate ranking compared with the baselines?

RQ3

How reliably does the quoted evidence support each inferred skill?

RQ4

Which kinds of errors does the LLM make?

ARCHITECTURE OVERVIEW

Seven stages, from data to analysis

1 · Data

Three datasets, official splits

2 · Preprocessing

Clean text, extract explicit skills

3A · Job side

Task B: encoders fine-tuned on D2, core and contextual weights

3B · Resume side

Task A: LLM infers skills with quotes, evidence check

4 · ESCO mapping

Both sides mapped to ESCO skills

5 · Matching

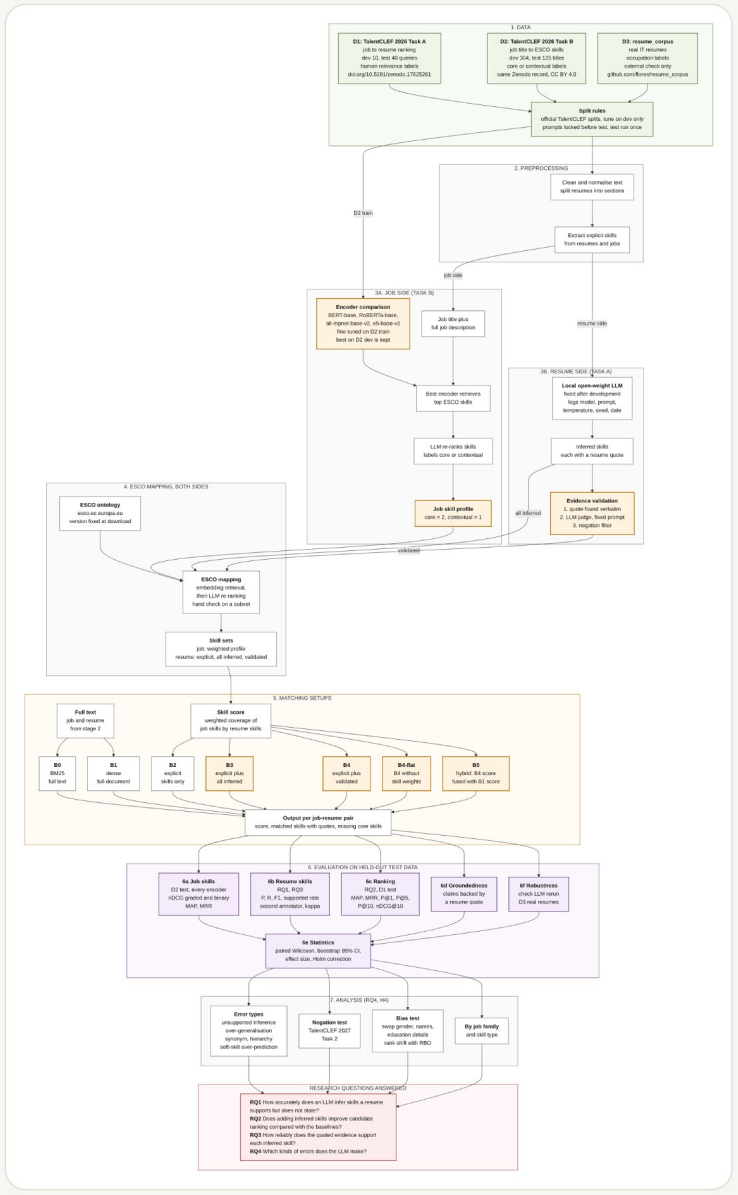
Seven setups, B0 to B5

6 · Evaluation

Held-out test data, significance tests

7 · Analysis

Error types, negation, bias



ARCHITECTURE DIAGRAM

The full pipeline on one page

Read top to bottom. The job side and the resume side run in parallel and meet at the ESCO mapping. Zoomed views of each part follow the slide on that stage.

- Green: data
- Orange: this project's own contribution
- Purple: evaluation
- Red: research questions answered

STAGE 1 · DATA

Two TalentCLEF datasets, one external check

D1 · Task A

TalentCLEF 2026, job to resume ranking

Dev 10, test 40 queries per language

Human relevance labels

D2 · Task B

TalentCLEF 2026, job title to ESCO skills

Dev 304, test 125 job titles

Core or contextual labels

D3 · resume_corpus

Real IT resumes with occupation labels

External check only

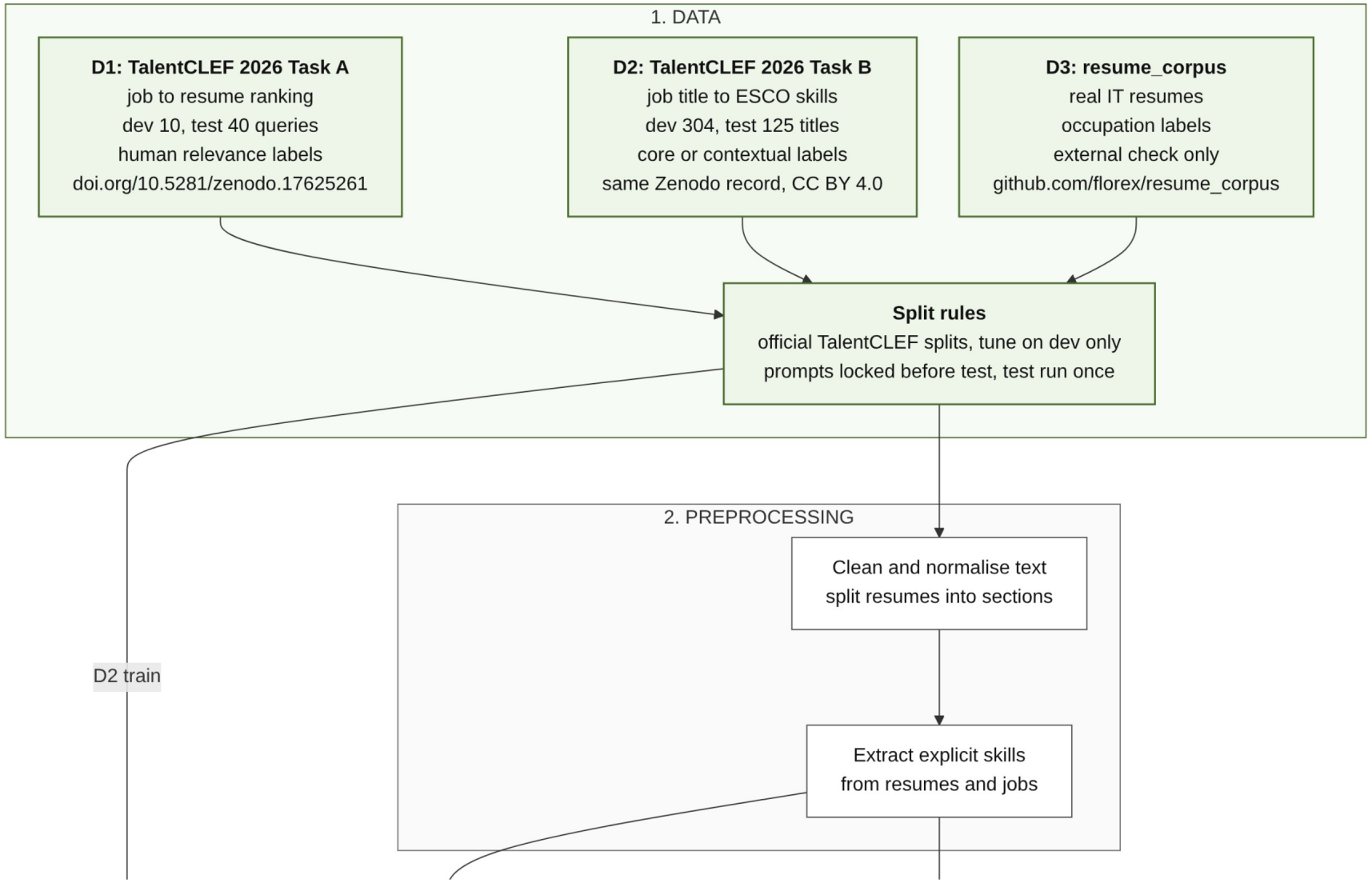
Licence and ethics to confirm

Split rules

Official TalentCLEF splits · tuning on dev only · prompts locked before test · test run once

Sources: Gasco et al. (2026); Zenodo 10.5281/zenodo.17625261 (CC BY 4.0); github.com/florex/resume_corpus · 7

DIAGRAM · STAGES 1 AND 2: DATA AND PREPROCESSING



STAGES 2 AND 3

Task B builds the job side, Task A the resume side

3A · Job side

Input: job title plus the full job description

Encoders fine-tuned on D2: BERT-base, RoBERTa-base, all-mpnet-base-v2, e5-base-v2; best on D2 dev kept

Retrieve and re-rank: top ESCO skills, LLM labels each core or contextual

Profile: core = 2, contextual = 1

3B · Resume side

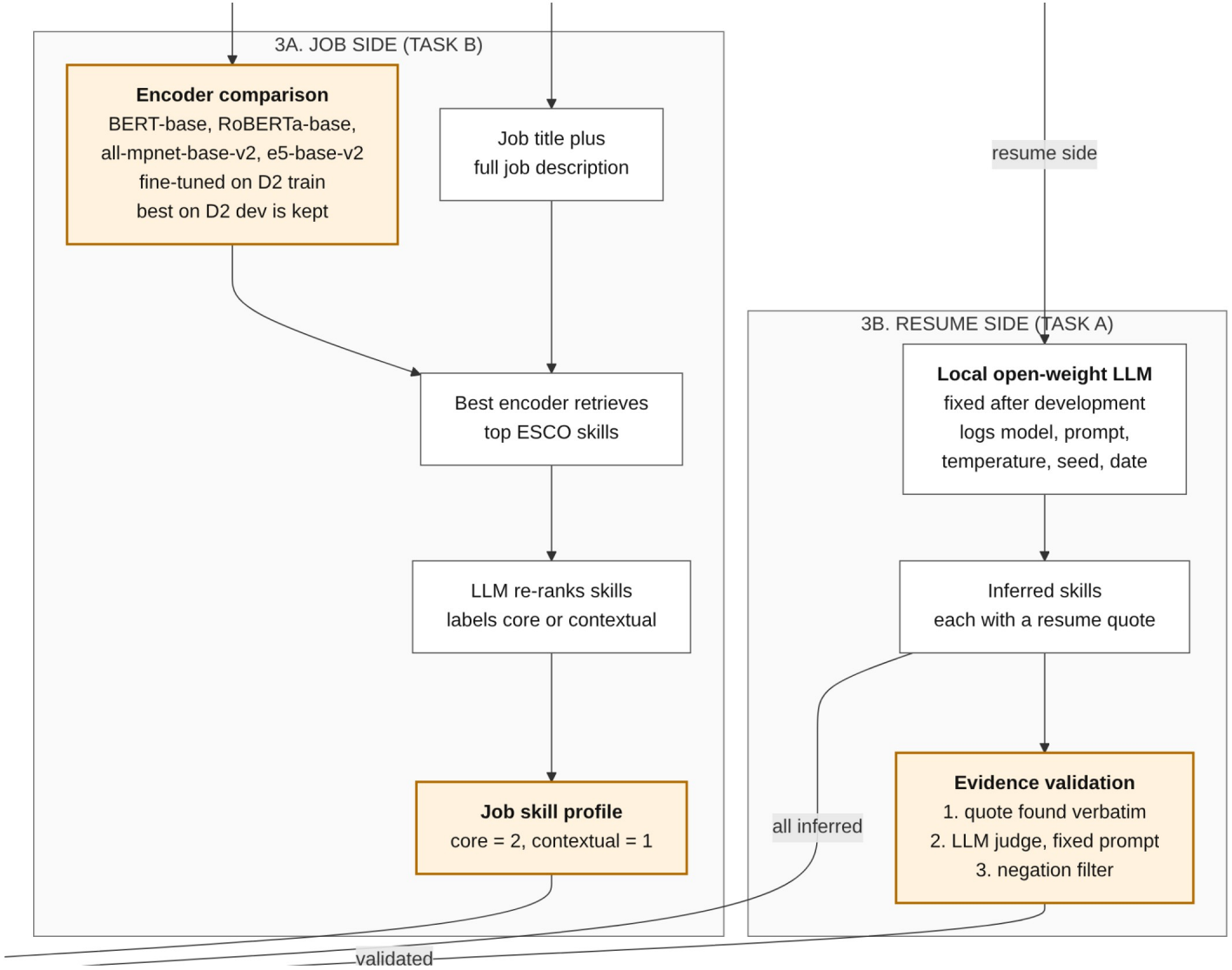
Local open-weight LLM: fixed after development

Inferred skills: each with a quoted resume sentence

Evidence check: quote found verbatim, LLM judge with a fixed prompt, negation filter

Logged per run: model version, prompt, temperature, seed, date

DIAGRAM · STAGE 3: JOB SIDE AND RESUME SIDE



STAGES 4 AND 5 · ESCO MAPPING AND MATCHING

Seven setups isolate each design choice

Setup	Representation	Role
B0	BM25 on full text	Baseline
B1	Dense full-document similarity	Baseline
B2	Explicit skills only	Baseline
B3	Explicit plus all inferred skills	No evidence check
B4	Explicit plus validated inferred skills	Main method
B4-flat	B4 without core and contextual weights	Tests Task B weights
B5	B4 score fused with B1 score	Hybrid

Main comparisons: B3 against B4, B4 against B4-flat, B5 against B1. ESCO mapping uses embedding retrieval, then LLM re-ranking (Clavié and Soulié, 2023).

DIAGRAM · STAGE 4: ESCO MAPPING, BOTH SIDES

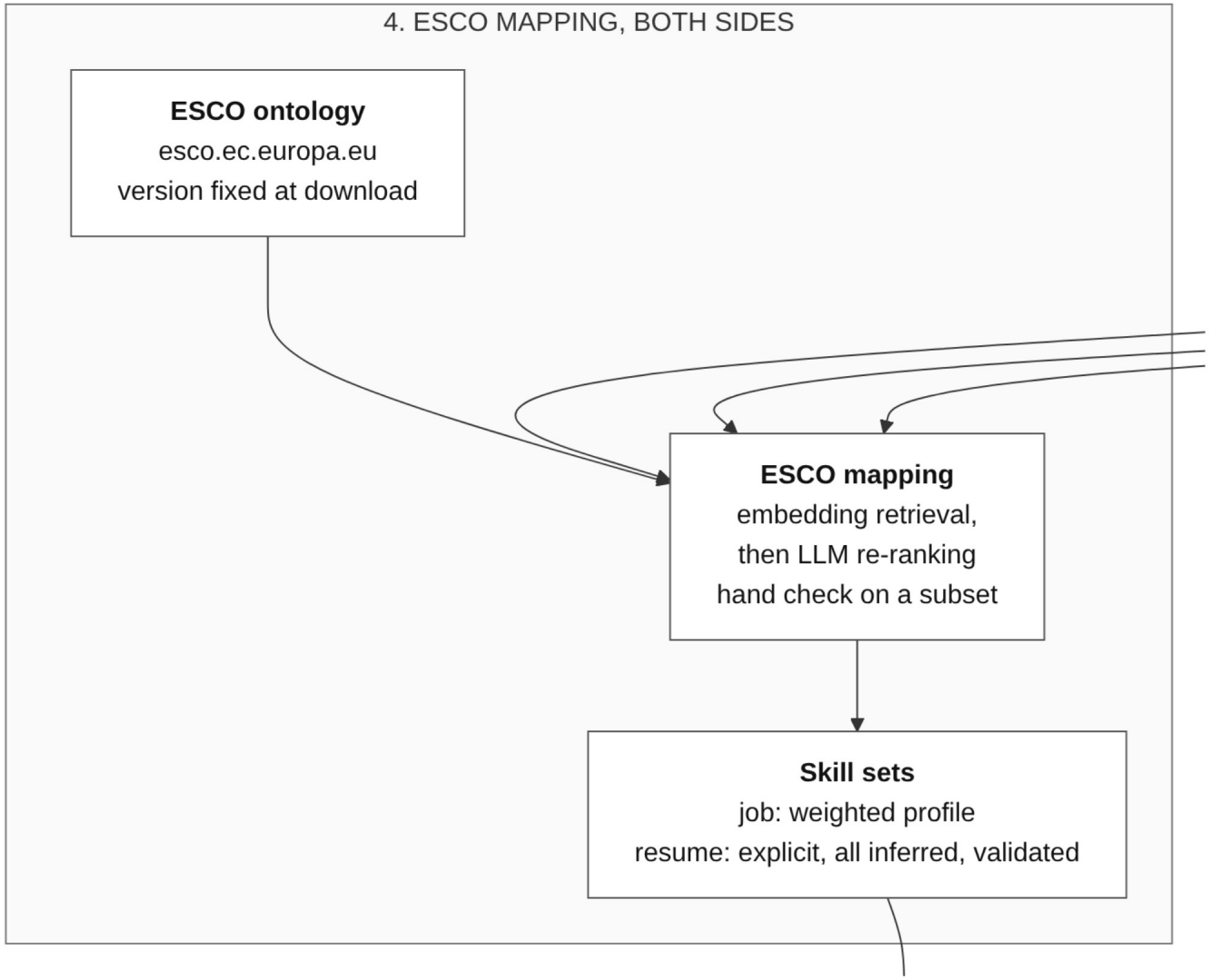
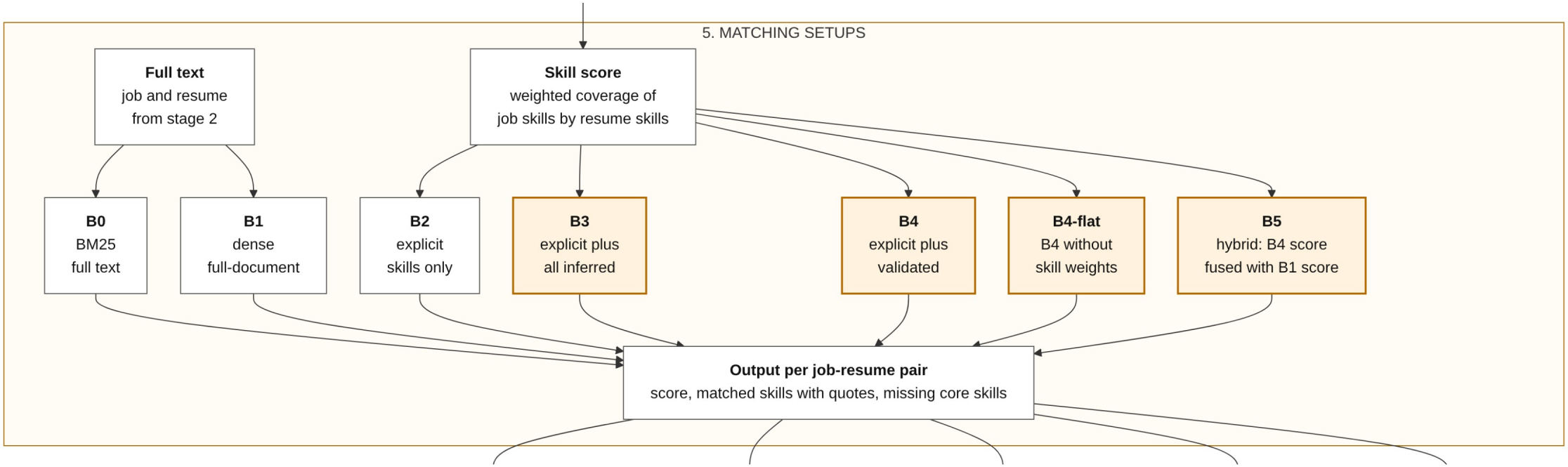


DIAGRAM · STAGE 5: MATCHING SETUPS



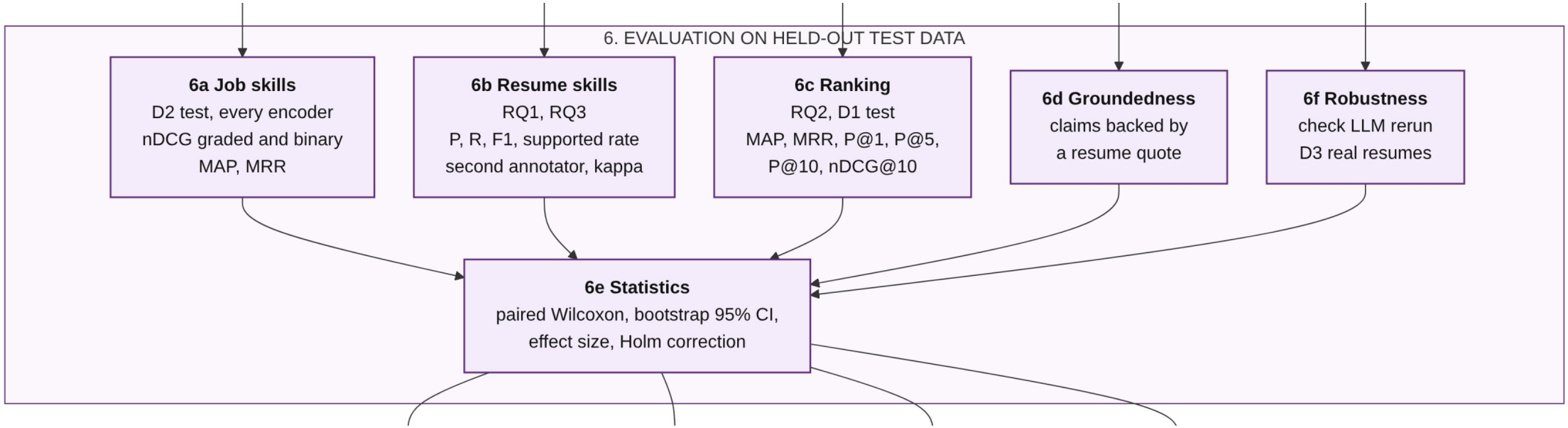
STAGE 6 · EVALUATION

Each part is scored on held-out test data

What	Data	Metrics
Job skill prediction	D2 test	nDCG graded and binary, MAP, MRR
Resume skills (RQ1, RQ3)	Annotated subset	Precision, recall, F1, supported rate, Cohen’s kappa
Candidate ranking (RQ2)	D1 test	MAP, MRR, P@1, P@5, P@10, nDCG@10
Groundedness	Ranking output	Share of claims backed by a resume quote
Robustness	D3, check LLM	B3 and B4 rerun
Statistics	Per query	Paired Wilcoxon, bootstrap 95% CI, effect size, Holm

Reference: TalentCLEF 2026 official baselines, Task A English MAP 0.4001, Task B graded nDCG 0.6204 · 14

DIAGRAM · STAGE 6: EVALUATION ON HELD-OUT TEST DATA



STAGE 7 · ANALYSIS (RQ4, H4)

Where the LLM goes wrong, and for whom

Error types

Unsupported inference, over-generalisation, synonym and hierarchy confusion, soft-skill over-prediction

Negation test

Skills the text denies, in line with TalentCLEF 2027 Task 2

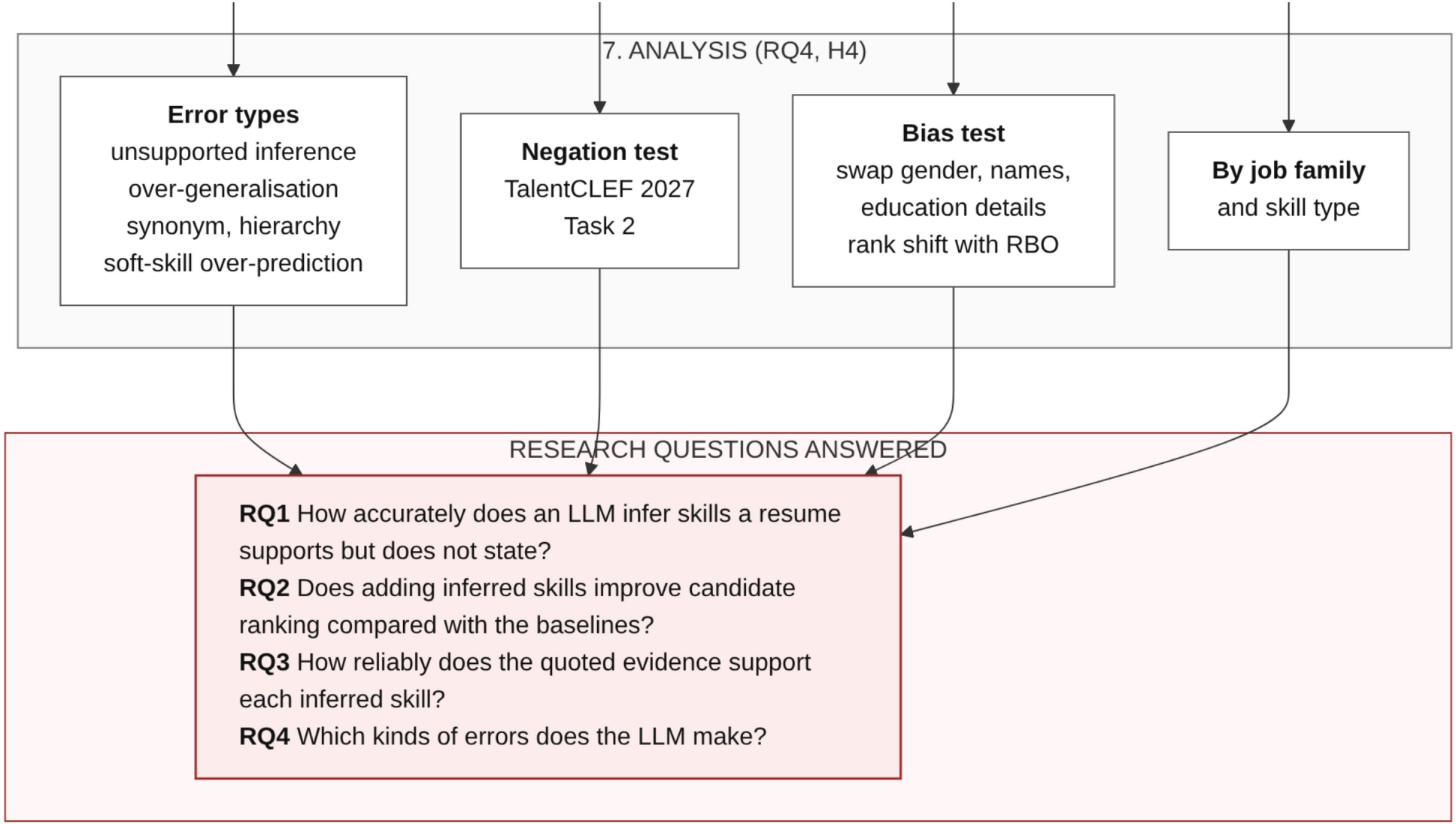
Bias test

Swap gender cues, names and education details, then measure rank shift with RBO (Iso et al., 2025)

By job family

Results split by job family and skill type to test H4

DIAGRAM · STAGE 7 AND THE RESEARCH QUESTIONS



WHAT THE PROJECT ADDS

Each gap in earlier work has a stage here

Source	Gap	This project
TalentCLEF 2026	No team fed Task B skills into Task A ranking	Task B builds the job profile used in Task A
TalentCLEF 2026	Task B input is a job title only	Adds the full job description
Vaishampayan et al. 2025	One LLM, observational, no control	Controlled B3 against B4 test, check LLM
Jiechieu, Tsopze 2021	IT resumes only, explanations not tied to text	Quoted evidence for every skill, ESCO labels
Goyal et al. 2023	Job side only	Joins the job side and resume side

PLAN

Five milestones to submission

2 to 6 Nov 2026

Interim meeting and interim report

End Dec 2026

Data and ethics settled, annotation guidelines, prototype, two baselines

End Jan 2027

Remaining setups built, training and testing done

End Feb 2027

Evaluation and failure analysis complete

20 Apr 2027

Dissertation, poster and evidence submitted, 3pm

KEY SOURCES

Papers behind the architecture

Gasco et al. 2026. Overview of the TalentCLEF 2026. arXiv:2606.31692

Vaishampayan et al. 2025. Human and LLM-Based Resume Matching. Findings of NAACL 2025

Jiechieu and Tsopze 2021. Skills prediction with CNN. Neural Comput. Appl.

Iso et al. 2025. Evaluating bias in LLMs for job-resume matching. NAACL 2025 Industry

Gasco et al. 2025. Overview of the TalentCLEF 2025. arXiv:2507.13275

Goyal et al. 2023. JobXMLC. Findings of EACL 2023

Clavié and Soulié 2023. LLMs as zero-shot ESCO skills matchers. arXiv:2307.03539

Gugnani and Misra 2020. Implicit skills extraction using document embedding. AAAI 2020